

The website tech stack will consist of HTML and CSS for the frontend and JavaScript running the backend. This stack allows us to easily run our applications on phones and computers, as web applications are very portable and the only thing that we will need to change is the CSS. We do not need to use frontend Javascript frameworks like React or Vue, as vanilla JavaScript is enough to handle our needs for now (this might change in the future). We also do not need to worry about using a special authentication library, as Firebase will handle account authentication and the database management. Users will listen to their music by uploading their song files to our web application and then playing the music for the MVP. For the final product, we will try to find a way to query and download songs. Using user files rather than platform integration saves us the complexity of managing platform-specific login information and API calls. Firebase offers a cloud storage bucket for handling the storage of users' songs, and all other data can be stored in a cloud SQL/NoSQL database. The database and bucket will be connected as each playlist in the database will hold a reference to its corresponding storage bucket. The cloud provider for our storage and database does not matter, there are several companies that offer what we need. We will hold the project in a GitHub repository, this provides us with version control and free website hosting via GitHub Pages. Our entire tech stack uses popular technologies and frameworks, making learning easy as there are a lot of resources available.